

Assignment 2 Case Study

Stage 2 Tutorial From Problems to Requirements

Week 5 | 60 minutes

Learning goals

- Analyse stakeholders.
- Distinguish functional and non-functional requirements.
- Recognise ambiguity and unsupported requirements.
- Define scope.
- Develop user stories and acceptance criteria.
- Critique AI-generated requirements.

Activity 1 - Stakeholder Map

Stakeholder	Need	Potential conflict
Patients	Easy and flexible appointment scheduling	Cancellation policies (especially cancellation on short notice)
Practitioners	Well organized daily schedules, preventing double bookings and adequate time for consults	Consults running over time
Receptionists	User friendly GUI to create and manage patient appointments	System could be counterintuitive or slow to use during peak times
Practice Management	Maximizing clinic's efficiency, accurate audit trails, compliance with health regulations and system reliability	Software development costs, scope creep, running over deadlines
Software Developers	Clear system requirements, maintainable code architecture, no scope creep	Inability to meet client deadlines

Activity 2 - Functional or Non-Functional?

- ☒ Functional ☐ Non-functional The system shall allow staff to cancel an appointment.
- ☐ Functional ☒ Non-functional The system should remain responsive for the course-scale dataset.
- ☒ Functional ☐ Non-functional The system shall retain cancelled appointments.
- ☐ Functional ☒ Non-functional Core business logic should be independently testable.
- ☒ Functional ☐ Non-functional The system shall search for a patient by ID.

Activity 3 - Repair Ambiguous Requirements

The system should be easy to use.

Problem: 'Easy to use' is vague and subjective Clarification question: What specific usability metric should we target?

Patient search should be fast.

Problem: 'Fast' is vague and subjective Clarification question: What is the maximum acceptable response time in seconds or milliseconds for a search query?

The system should securely manage data.

Problem: 'Securely' is an abstract term Clarification question: What specific security standards or regulations must the data handling comply with?"

Appointments should normally be easy to cancel.

Problem: 'normally' and 'easy' are both terms open to interpretation Clarification question: What constitutes a 'normal' cancellation versus an exception or edge case?

Activity 4 - AI Requirements Audit

Classify each suggestion: Confirmed / Assumption requiring validation / Unsupported / Out of scope.

AI suggestion	Classification	Evidence / reason
Patients receive SMS reminders.	Assumption requiring validation	SMS notifications are a common feature for medical bookings, but sending external SMS alerts is not explicitly defined in the core requirements.
Facial recognition login.	Out of scope	Biometric authentication adds significant hardware dependency and complexity.
Receptionists create appointments.	Confirmed	Directly matches the core business model where receptionists act as the primary administrative users scheduling bookings for patients with practitioners.
Online payment.	Out of scope	Payment processing, payment gateway integration, and invoicing are out of scope for the core booking engine.
Practitioners view schedules.	Confirmed	Directly supported by requirements enabling practitioners to view their allocated daily appointment slots.
AI recommends treatments.	Unsupported	Clinical decision support and AI medical diagnostics are completely absent from the project scope
Cancelled appointments remain in	Confirmed	Directly satisfies requirements

history.		mandating that cancelled appointments are preserved as historical objects rather than hard-deleted.
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Exit question

Why is 'AI suggested it' not sufficient evidence for a requirement?

Generative AI models lack context regarding real-world business constraints, clinical safety standards, and user needs.